Cretaceous Coast Belt paleomagnetic data from the Spetch Creek pluton, British Columbia: evidence for the "tilt and moderate displacement" model[1]

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Well-grouped stable magnetizations have been isolated at 14 of 20 sites sampled from the Spetch Creek pluton. The single-polarity primary magnetization directed at $D = 026.2°$, $I = 73.5°$ ($\alpha_{95} = 4.4°$, paleolatitude 59 ± 7°N, paleopole 73°N, 078°W, $A_{95} = 7°$) was acquired around 88 Ma during the Cretaceous normal polarity superchron (118–84 Ma). This direction is discordant from the expected mid-Cretaceous direction ($D = 332.5°$, $I = 75.1°$) for North America. The difference could be caused by one of two end member models: the all-tilt model requires a 15° east-side-up tilt about a horizontal axis striking 353°, and the displacement–rotation model requires 330 ± 770 km of northward displacement combined with 54 ± 14° of clockwise rotation. Regardless, this result provides a negative test of the Baja British Columbia model, which requires ~2400 km of northward displacement.

A review of previously observed mid-Cretaceous magnetizations from the Coast Belt, which are also discordant, indicates that they exhibit common characteristics, although their discordance is not uniform. Assuming that present horizontal approximates paleohorizontal, post-mid-Cretaceous latitudinal displacements inferred from individual results vary between 330 and 3500 km northwards. Relative rotations about vertical axes vary between 17 and 57° clockwise. Such variation cannot be accounted for by the displacement and rotation of a superterrane as a whole. Recent studies emphasize that many of the intrusions have probably been locally tilted. Consistent with known geology, these discordant poles are best explained by the "tilt and moderate displacement" model. This model invokes moderate (500–1000 km) post-mid-Cretaceous northward displacement of the amalgamated Insular, Coast, and Intermontane terranes west of the major dextral Canadian Cordilleran fault systems, combined with variable local block tilting east- and northeast-side-up. Northward displacement was driven by Kula and (or) Farallon – North American plate interactions from 90 to 56 Ma. Tilting is most likely due to a combination of northeast–southwest compression, differential uplift, and extension.

Des aimantations stables cohérentes ont été isolées pour 14 des 20 sites échantillonnés dans le pluton de Spetch Creek. L'aimantation de polarité primaire simple de direction $D = 026{,}2°$, $I = 73{,}5°$ ($\alpha_{95} = 4{,}4°$, paléolatitude 59 ± 7°N, paléopôle 73°N, 078°W, $A_{95} = 7°$) fut acquise autour de 88 Ma, durant la superchrone crétacée de polarité normale (118–84 Ma). Cette direction est discordante de celle appréhendée pour le Crétacé moyen ($D = 332{,}5°$, $I = 75{,}1°$) de l'Amérique du Nord. Cette différence pourrait être expliquée par un de deux modèles opposés : soit par le modèle fondé entièrement sur un basculement, lequel requiert un soulèvement du côté oriental du bloc dû à une rotation de 15° autour d'un axe horizontal de direction 353°, et par le modèle de déplacement–rotation impliquant une dérive de 330 ± 770 km vers le nord combinée avec une rotation de 54 ± 14° dans le sens horaire. Quoi qu'il en soit, les résultats obtenus ici sont en désaccord avec le modèle de Baja British Columbia qui nécessite un déplacement vers le nord de ~2400 km.

L'examen des aimantations connus pour le Crétacé moyen du Domaine côtier, étant elles aussi discordantes, indique qu'elles possèdent des caractères communs en dépit de leurs discordances non uniformes. En supposant que les composantes horizontale actuelle et paléohorizontale sont aussi approchées que possible, les déplacements latitudinaux postérieurs au Crétacé moyen, estimés à partir de résultats individuels, varient de 330 à 3500 km vers le nord. Les rotations relatives autour des axes verticaux varient de 17 à 57° dans le sens horaire. Une telle variation ne peut pas être expliquée par le déplacement et la rotation d'un superterrane se comportant comme une entité. Les études récentes appuient sur le fait que plusieurs des intrusions ont été dérangées probablement par un basculement local. L'explication de ces pôles discordants, celle qui s'accorde le mieux avec les connaissances géologiques acquises, est fondée sur le modèle qui implique « un basculement et un déplacement modéré ». Ce modèle invoque un déplacement modéré (500–1000 km) vers le nord, postérieurement au Crétacé moyen, des terranes amalgamés des Domaines insulaire, côtier, et intermontagneux, à l'ouest des systèmes de failles dextres majeurs de la Cordillère canadienne, combiné à un basculement local et variable de blocs soulevés du côté est et nord-est. Le déplacement vers le nord a été produit par une poussée créée par les interactions de la plaque de Kula et (ou) Farallon avec la plaque de l'Amérique du Nord entre 90 et 56 Ma. Le basculement résulte probablement des effets combinés d'une compression nord-est – sud-ouest, et d'une extension et soulèvement différentiels.

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Previous Cretaceous Coast Belt paleomagnetic studies

The development of a large-scale northward displacement and rotation model, progressively refined into the Baja British Columbia model (e.g., Beck 1976; Irving 1985; Irving et al. 1985; Umhoefer 1987) for part of the western Cordillera, began with Beck and Noson's (1972) interpretation of discordant paleomagnetic directions observed in rocks from the mid-Cretaceous Mount Stuart batholith (MS, Fig. 1) in the northern Cascade Range. They suggested that the discordant shallow inclinations from Mount Stuart were acquired by the batholith in a mid-Cretaceous latitude over 3000 km to the south of its present position relative to North America, and that tilt of such a large batholith was unlikely. Shortly thereafter, Symons (1973*a*) tested their large-scale displacement hypothesis in a paleomagnetic study of mid-Cretaceous intrusive rocks from Howe Sound (HS, Fig. 1) in the southwestern Coast Belt. He provided a larger data set (17 sites) than the one for the Mount Stuart batholith (3 sites). Paleomagnetic directions from intrusives in Howe Sound were also discordant; however, the discordance was much smaller than that observed

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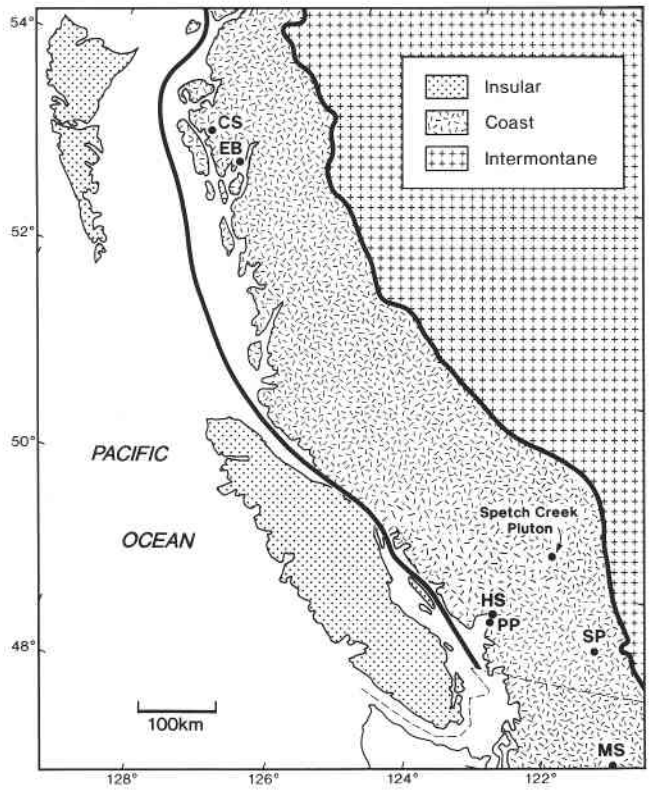


FIG. 1. General tectonic setting of the Spetch Creek pluton within the southern Coast Belt and previous Cretaceous paleomagnetic studies. Note the northern Casacade Range has been correlated with the Coast Belt. Localities as follows: MS, Mount Stuart batholith; HS, Howe Sound intrusives; EB, Ecstall and Butedale plutons; CS, Captains Cove and Stephen Island plutons; SP, Spuzzum pluton; PP, Porteau pluton. Belt boundaries from Wheeler et al. (1991).

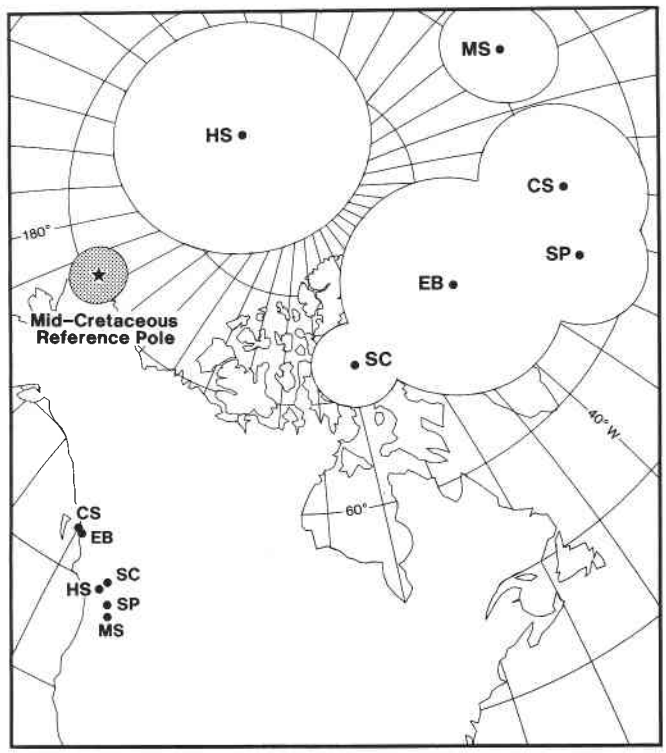


FIG. 2. Plot of Cretaceous Coast Belt pole positions and their A_{95} errors. Pole symbols match localities from Fig. 1. SC is the Spetch Creek pole from this study and the star is the position of the North American mid-Cretaceous reference pole (Globerman and Irving 1988).

from the Mount Stuart batholith. Thus if you accept, as Beck and Noson (1972) did, that paleohorizontal for the intrusive body is the same as present horizontal, then the Howe Sound data indicate that these Cretaceous intrusive rocks could not have been magnetized thousands of kilometres to the south. Alternatively, Symons (1973*a*) argued that the discordance of the Howe Sound paleopole with the expected Cretaceous pole was best explained on the basis of geologic evidence by horizontal axis tilt ~15° down to the southwest, without latitudinal displacement.

Thus began the "tilt" versus "displacement" controversy, which has gone unresolved for the last two decades. The following synopsis of subsequent paleomagnetic studies of rocks in the Coast Belt (CB) sets the framework for a reassessment of the tilt versus displacement discussion in light of additional data from the mid-Cretaceous Spetch Creek pluton (Fig. 1).

Work by Symons (1974, 1977*a*) on the Cretaceous Ecstall and Butedale (EB) plutons in the central part of the CB (Fig. 1) provided similar paleomagnetic directions, which he combined. The resulting pole (EB) is removed from the Mount Stuart (MS), Howe Sound (HS), and Cretaceous reference poles (Fig. 2.). Symons (1977*a*) concluded that the discordant pole was the result of postemplacement tilt of ~15° down to the southwest, citing field evidence from structural trends, from fault, fold, contact, and foliation attitudes, and from the distribution of plutonic phases. The emphasis of his study was to provide a regional-scale fold test using the Hawksbury warp that demonstrated that Paleogene folding of part of these plutons had taken place. The paleomagnetic evidence for this folding was recorded in both Cretaceous (Ecstall and Butedale) and Eocene (Ponder, Alastair Lake, and Quottoon) plutons over a large area of the central CB. Thus, Symons (1977*a*) rejected the large-scale northward translation and clockwise rotation models of Beck and Noson (1972) and Beck (1976) because these mechanisms could not account for the variation observed in discordant inclinations and declinations. Symons (1977*b*) also observed discordant paleomagnetic data from the nearby Eastern Panel of the western age zone of the CB (84–140 Ma, Hutchison 1967) comprised of the Captain Cove, Stephens Island, and Gil Island plutons. Obtaining a discordant direction, Symons (1977*b*) similarly interpreted that the Eastern Panel was tilted ~30° down to the southwest, as evidenced by the attitudes of contacts and foliations, plutonic trend directions, distribution of metamorphic grades, and the consistent data from the EB plutons. The position of the Eastern Panel pole (CS) is also removed from other CB poles (Fig. 2). This variation in pole positions again suggested to Symons (1977*b*) that northward translation and clockwise rotation models could not account for this directional variability because the discordant effect should be uniform and this was not the case (see poles MS, HS, EB, CS in Fig. 2).

Subsequent work by Beck et al. (1981) provided a much larger paleomagnetic data set for the Mount Stuart batholith but did not substantially change the discordant direction (Beck and Noson 1972). Beck et al. (1981) considered both the tilt and translation models to account for the discordance, and again they favoured large-scale northward translation and

clockwise rotation, suggesting that the local geology did not support tilt. Given this unresolved tilt versus displacement discussion, Irving et al. (1985) carried out a paleomagnetic study of the mid-Cretaceous Spuzzum and Porteau plutons (SP and PP, Fig. 1) of the southern CB directed at resolving the controversy. They identified two magnetic components, A and B. The primary normal-polarity A component is considered to have been acquired during emplacement in the mid-Cretaceous. The normal-polarity B component is considered to be a secondary Tertiary component. The A component pole (SP) is discordant and similar in position to the Eastern Panel pole of Symons (1977*b*) (CS, Fig. 2). Irving et al. (1985) reasoned, as did Beck (1980), that Canadian Cordilleran (CC) Cretaceous poles were systematically displaced from the mid-Cretaceous reference pole. Their analysis required the omission of the Howe Sound pole (HS, Fig. 2), which Irving et al. (1985) rejected on the grounds that its position was inconsistent with that of the rest of the Cretaceous poles. They argued that, given their evidence of a secondary B component magnetization, similar in direction, the Howe Sound pole is probably representative of a Tertiary overprint magnetization. So selected, the Cretaceous data consistently exhibited flattened inclinations and large clockwise rotations (Fig. 14 of Irving et al. 1985). In their discussion Irving et al. (1985) included four poles from outside the CB. Three of these poles were derived from suspected overprint magnetizations of unknown age, although they were interpreted to be mid-Cretaceous on the basis of positional agreement. The position of the fourth pole from the Axelgold intrusion was uncertain because of a paleohorizontal ambiguity. Nevertheless, given the systematic discordance of these poles from such a large region, Irving et al. (1985) argued that this could not be the result of uniform tilt. Their favoured interpretation is recognized as the Baja British Columbia model, a refined version of the model discussed by Beck (1976) in which the amalgamated Insular, Coast, and Intermontane belts were driven northward by 2400 km following the mid-Cretaceous and underwent $\sim 60°$ of clockwise rotation before final collision and amalgamation with the North American margin. Like previously discussed large-scale displacement (Beck 1976; Beck et al. 1981), this model is dependent on the fundamental assertion that present horizontal approximates paleohorizontal for all the studied CB plutons.

Current model constraints

Recently, Butler et al. (1989) and Vandall (1990) have argued that several of the poles used to demonstrate the systematic discordance of mid-Cretaceous data by Irving et al. (1985) are not mid-Cretaceous in age. For example, there is no evidence that magnetizations from the Early Jurassic Guichon (Symons 1971, 1983) and Copper Mountain intrusives (Symons 1973*b*; Symons and Litalien 1984) are mid-Cretaceous overprints, as both plutons have K–Ar dates of ~ 200 Ma, normal and reverse polarities, paleolatitudes in agreement with those from Early Jurassic (~ 200 Ma) CC studies, and no evidence of thermal overprints. Thus, several poles should not have been part of the discussion.

It is now understood that the major elements of the CC were all in their present position relative to North America by the mid-Eocene based on concordant paleomagnetic data from bedded Eocene rocks (e.g., Symons and Welling 1989; Vandall and Palmer 1990*a*). Because of this, discordant magnetizations in Eocene and younger rocks cannot have been caused by latitudinal displacement, only by tilting and (or) rotations. Thus, the Eocene results from the Hawksbury warp (Symons 1977*a*) unequivocally document post-Eocene southwest down tilt and rotation in the CB. As a result, this supports Symons (1977*b*) conclusion that the adjacent Cretaceous plutons of the Eastern Panel (Captians Cove, Stephan Island, and Gil Island) have been similarly tilted.

Numerous recent studies from outside the CB also document paleomagnetically determined mid-Eocene tilts (e.g., Irving and Archibald 1990, in the Omineca Belt; Irving and Massey 1990, in the Insular Belt). Combined, these observations of discordant poles document widespread young pluton tilting and thereby demonstrate that paleohorizontal cannot be assumed to be the same as present horizontal when a pluton yields a discordant magnetization. Further, post-mid-Cretaceous faulting and tilting of domains in the CB are known to have occurred on the basis of abundant structural evidence (e.g., Rusmore and Woodsworth 1991; Monger and Journeay 1992; Varsek et al. 1992). This has undoubtedly tilted many older plutons, consistent with the geologic evidence of plutonic tilts that Symons (1973*a*, 1974, 1977*a*, 1977*b*) and Butler et al. (1989, 1990) have outlined. In addition, paleobarometric evidence from the Spuzzum pluton suggests that its discordant magnetization is the result of a southwest down tilt (Brown and Burmester 1991). By implication, it is fundamental to our understanding of CB tectonics that we acknowledge that present horizontal does not approximate paleohorizontal for all the studied Cretaceous CB plutons, as has been suggested (e.g., Symons 1977*a*; May and Butler 1986; Butler et al. 1989). In contrast, some plutons are not tilted and present horizontal approximates paleohorizontal, although this has only been demonstrated for plutons yielding a concordant magnetization (e.g., Irving and Archibald 1990; Vandall and Wheadon 1992). Thus the basis for discussing tilt, albeit with imperfections, has been considerably advanced as a result of numerous recent studies.

During much of the period in which these Cretaceous CB intrusions were studied, a mobilistic view for parts of the western Cordillera was supported by paleomagnetic data from the Triassic Karmutsen Formation and Takla Group and the Jurassic Hazelton Group. These data from bedded rocks were interpreted as indicating large (>1000 km) latitudinal displacements from the south (e.g., Irving and Yole 1972; Schwarz et al. 1980; Yole and Irving 1980; Monger and Irving 1980). As a result, the Baja British Columbia model fits well with this mobilistic view. More recent work has shown, however, that the original paleolatitudes calculated from these data were in error due to the the use of incorrect reference poles. The first suggestion of these errors was made by Gordon et al. (1984) and subsequently May and Butler (1986), and Vandall and Palmer (1990*b*) with additional data from the Hazelton Group, demonstrated that these data are in fact latitudinally concordant with updated reference poles. This indicates that when these rocks formed during the Triassic and Jurassic they were situated in much the same latitudinal position relative to North America as they are now. On the basis of the latitudinally concordant pre- and post-Cretaceous paleomagnetic data, and mid-Cretaceous to Eocene cumulative dextral offsets along major CC faults between 500 and 900+ km (Gabrielse 1985; Price and Carmichael 1986), Vandall (1990) concluded that large-scale (2400 km) post-Cretaceous northward latitudinal transport is unlikely. Following a review of the CC paleo-

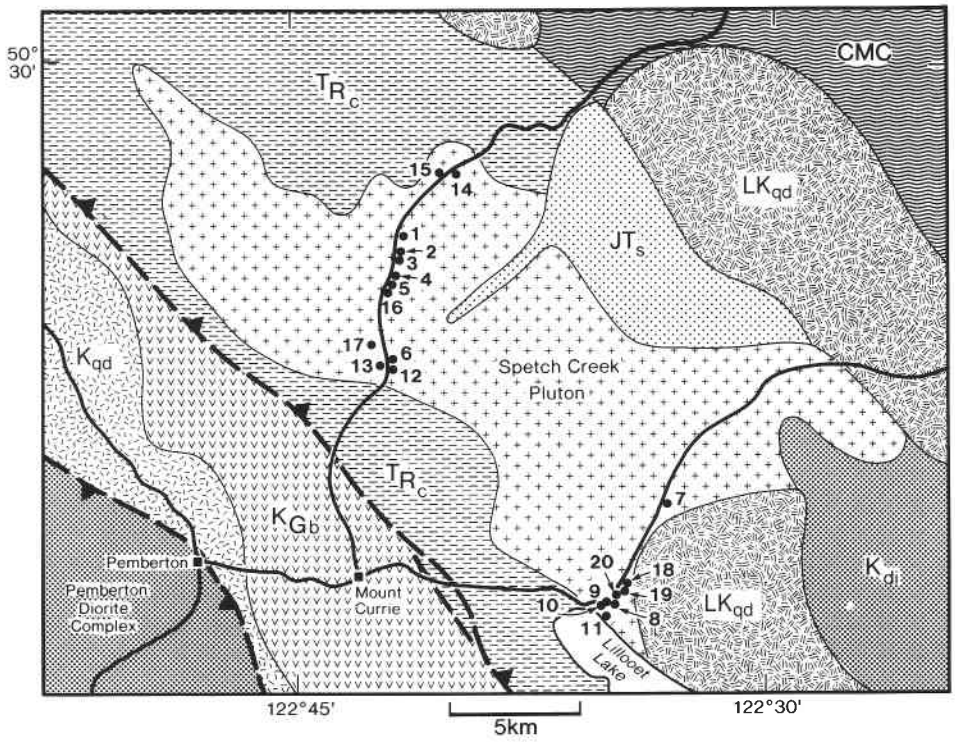


FIG. 3. Spetch Creek pluton paleomagnetic site locations. Geology after Journeay (1990). Map symbols as follows: LK_{qd}, Late Cretaceous quartz diorite; K_{di}, Cretaceous diorites; K_{qd}, Cretaceous quartz diorite; K_{Gb}, Cretaceous Gambier Group; JT_s, Jurassic Tyaughton Group; T_{R_c}, Triassic Cadwallader terrane rocks; CMC, Cascade metamorphic core. Heavy broken lines are major thrust faults.

magnetic data, Vandall (1990) proposed the "tilt and moderate displacement" (TMD) model as the most consistent fit to the geologic models for the assembly of the CC (e.g., Monger et al. 1982; van der Heyden 1992). In the TMD model the observed discordant Cretaceous poles are the result of post-mid-Cretaceous moderate northward displacement of 1000 km combined with concurrent variable local block tilting for amalgamated CC terranes west of the major dextral CC fault systems. Thus recent work has considerably advanced the discussion of tilt and displacement, which herein is evaluated with respect to new data from the mid-Cretaceous Spetch Creek pluton in the CB.

Geology and sampling

The Spetch Creek pluton is a mid-Cretaceous granodiorite intrusion that is situated in the southern CB (Fig. 1). Its crystallization age from a U–Pb zircon date is 102 ± 1 Ma (Friedman 1990). The pluton extends 33 km along its northwest axis and up to 16 km along its northeast axis (Fig. 3). The bulk of the pluton intrudes Upper Triassic island-arc volcanic rocks of the Cadwallader terrane (Fig. 3), which is interpreted to be allochthonous with respect to North America, and was amalgamated with the adjacent Bridge River oceanic terrane in the Middle Jurassic (Rusmore et al. 1988). Immediately west of the pluton and Cadwallader terrane rocks are major faults of the imbricate Coast Belt Thrust System, which separates the Western Coast Belt from the Eastern Coast Belt (Monger and Journeay 1992). Here high-angle reverse faults juxtapose rocks of the Cadwallader terrane against those of the Cretaceous Gambier Group with displacements in northeast-side-up sense (Fig. 3) (Journey 1990). The timing of this southwest-vergent thrusting and associated crustal shortening is bracketed between 97 and 91 Ma (Monger and Journeay 1992) and appears to have been synkinematic with uplift and cooling of the Spetch Creek pluton. Following this, between 91 and 86 Ma, northeast-vergent thrusting occurred, as shown by the southwest-dipping thrust faults in the western Bendor Range (Monger and Journeay 1992). These structures extend to depth, cutting southwest-vergent thrusts, and may have been active at deeper crustal levels over a longer period of time (Monger and Journeay 1992; Varsek et al. 1992). Conventional K–Ar cooling ages on biotite of 86.2 ± 3.1 Ma (Wanless et al. 1979) and 89.8 ± 3.4 Ma (Stevens et al. 1982) suggest that the pluton's primary magnetization was acquired after southwest-vergent thrusting. The massive, unfoliated, and unfaulted character of the Spetch Creek pluton supports this conclusion as does the observed magnetization direction, which is discussed later. Discordance between biotite K–Ar ages and U–Pb zircon ages is a common feature in the CB, and likely reflects relatively slow pluton cooling. The accuracy of an anomalously young hornblende age of 76.7 ± 3.9 Ma (Wanless et al. 1979) is believed to be poor, as it is likely the result of argon loss. It is much easier to disturb hornblende's isotope systematics, with only ~0.7% potassium, than biotite's, with ~7.0% potassium.

The Spetch Creek pluton consists of a clean, homogeneous, medium- to coarse-grained granodiorite, which is massive except near Lillooet Lake where it is foliated parallel with the shore (Price et al. 1985). Sampling of the pluton was carried out at 20 sites in different phases of the pluton along the Duffy Lake road beside Lillooet Lake and the Birken road going north of Mount Currie (Fig. 3). Site 12 is the 102 Ma U–Pb age date locality, sites 8–10 are proximal to the 86 Ma K–Ar age date locality, and site 1 is southwest of the 90 Ma K–Ar age date locality. The sampled rock types range in composition from fine-grained diorite xenoliths to medium-grained

TABLE 1. Site mean remanence data

Site	Rock type	UTM coordinates E	UTM coordinates N	M (A/m)	N	D (°)	I (°)	k	R	α_{95} (°)	VGP λ (°N)	VGP ϕ (°E)
1	gdi	522000	5587300	0.251	4	042	62	91	3.967	9.7	61	327
2	fdi−gdi	522000	5586300	0.173	—	—	—	—	—	—	—	—
3	fdi−gdi	522000	5586200	0.100	—	—	—	—	—	—	—	—
4	gdi	522300	5585450	0.133	4	083	68	31	3.904	16.7*	—	—
5	fdi	522400	5585300	0.187	5	039	62	44	4.910	11.6	63	328
6	di	522650	5582700	0.210	6	026	76	30	5.835	12.4	71	274
7	fdi	532300	5575700	0.114	5	044	61	111	4.964	7.3	59	326
8	di−fdi	530350	5572300	0.465	3	049	75	26	2.924	24.5*	—	—
9	fdi	529700	5572500	0.473	4	017	78	589	3.995	3.8	71	258
10	gdi−fdi	529650	5571950	0.818	4	016	83	88	3.966	9.9	63	245
11	gdi	529500	5571900	0.318	5	357	77	262	4.985	4.7	76	231
12	gdi	521300	5582600	0.061	5	012	75	149	4.973	6.3	77	262
13	gdi	521300	5582600	0.193	4	026	75	80	3.962	10.4	72	279
14	gdi	524250	5590350	0.337	4	030	78	177	3.983	6.9	68	270
15	gdi	524100	5590400	0.591	5	021	78	331	4.988	4.2	71	262
16	fdi	521500	5585200	0.131	5	027	61	59	4.933	10.0	69	342
17	gdi	520800	5583500	0.612	5	000	80	210	4.981	5.3	70	238
18	gdi	531600	5573300	0.054	2	009	77	540	1.998	10.8	74	250
19	gdi	531200	5573200	0.118	—	—	—	—	—	—	—	—
20	gdi	530600	5572950	0.089	—	—	—	—	—	—	—	—

NOTES: Rock types as follows: di, diorite; fdi, fine-grained diorite; gdi, granodiorite. M is the average NRM intensity of magnetization; N is the number of cores; D and I are the site mean declination and inclination of the magnetization given with respect to present horizontal; k is the best estimate of Fisher's (1953) precision parameter; R is the length of the resultant vector; α_{95} is the radius of the circle of 95% confidence; and λ and ϕ are latitude and longitude of the virtual geomagnetic pole (VGP). * indicates rejected sites on the basis of $\alpha_{95} > 15°$.

granodiorites (Table 1). Six thin and 11 polished thin sections representative of the range of lithologies and textures exhibited by the paleomagnetic cores were studied. Overall these rocks are very fresh and unfoliated, preserving primary textures. The abundant opaque minerals are mostly homogeneous magnetite grains ranging from 30 μm to submicron sizes. Hematite is noticeably absent except for trace amounts in two thin sections where it occurs as reaction rims around magnetite. Mineral assemblages include plagioclase, hornblende, biotite, quartz, and orthoclase feldspar. Accessory minerals, in decreasing order of abundance, include magnetite, ilmenite, rutile, sphene, chlorite, hematite, and epidote. Magnetite is present in all cases.

Methods

At each of 20 sites, five to seven cores were drilled and then oriented in situ using both sun and magnetic compasses, giving readings in good agreement. Two specimens were sliced from most cores. Their natural remanent magnetization (NRM) was analyzed using Robotic Systems International (RSI) automated Schonstedt SSM-1 and SSM-2 spinner magnetometers, a Schonstedt TSD-1 thermal demagnetizer, and a Sapphire Instruments SI-4 alternating-field (AF) demagnetizer.

After its NRM was measured, each specimen was subjected to AF or thermal step demagnetization in 8−12 steps. AF demagnetization routinely included: 5, 10, 15, 20, 30, 40, 60, 80, and 100 mT steps. Thermal demagnetization routinely included 200, 300, 400, 500, 550, 575, 600, and 650°C steps. Stable remanence directions in each specimen were obtained using stereographic, intensity decay and orthogonal demagnetization diagrams and least squares principal component analysis (Kirschvink 1980). Magnetizations were identified as lines on orthogonal plots (Zijderveld 1968; Kirschvink 1980) or as stable end points (McElhinny and Gough 1963). Directions of remanence were accepted with a maximum angle of deviation (MAD) of $< 10°$. Site mean remanence directions were calculated using two-tiered statistics (Fisher 1953; Watson and Irving 1957). Tectonic analysis of the relative rotations (RR's) (RR's $<$ 0 indicate clockwise rotation) and the relative paleolatitudinal displacements (RPD's) (RPD's > 0 indicate northward paleolatitudinal displacement) with respect to cratonic North America were calculated by the method of Yole and Irving (1980, 1981), incorporating the error analysis of Demarest (1983).

Results

Following detailed AF and thermal step demagnetization, 30% of the specimens exhibited unstable magnetizations and were rejected. All specimens from sites 2, 3, 19, and 20 (Table 1; Fig. 3) as well as specimens from 10 additional cores (sites 4, 8, 9, 10, 13, 14, and 18; Table 1) exhibited unstable magnetizations. Stable specimens from the remaining 14 sites responded well to progressive AF and thermal step demagnetization, yielding only normal polarity magnetizations. Many specimens exhibited the initial removal of low coercivity or low unblocking temperature (T_{UB}) magnetizations, sometimes directed along the present Earth's magnetic field (PEMF) and sometimes not, followed by linear univectoral decay (Figs. 4A, 5A). Most stable specimens were characterized by only linear univectoral decay, exhibiting a spectrum of coercivities and unblocking temperatures (Figs. 4B−4D, 5B−5E) but all broadly similar. The early-removed magnetizations that are aligned along the PEMF are interpreted as recent Bruhnes overprints, whereas those randomly directed are believed to be laboratory viscous remanence components.

Coercivity spectra are variable but all specimens exhibit the rapid removal of their remanence between 0 and 40 mT (Fig. 4). By 100 mT the magnetic remanence of all specimens had been

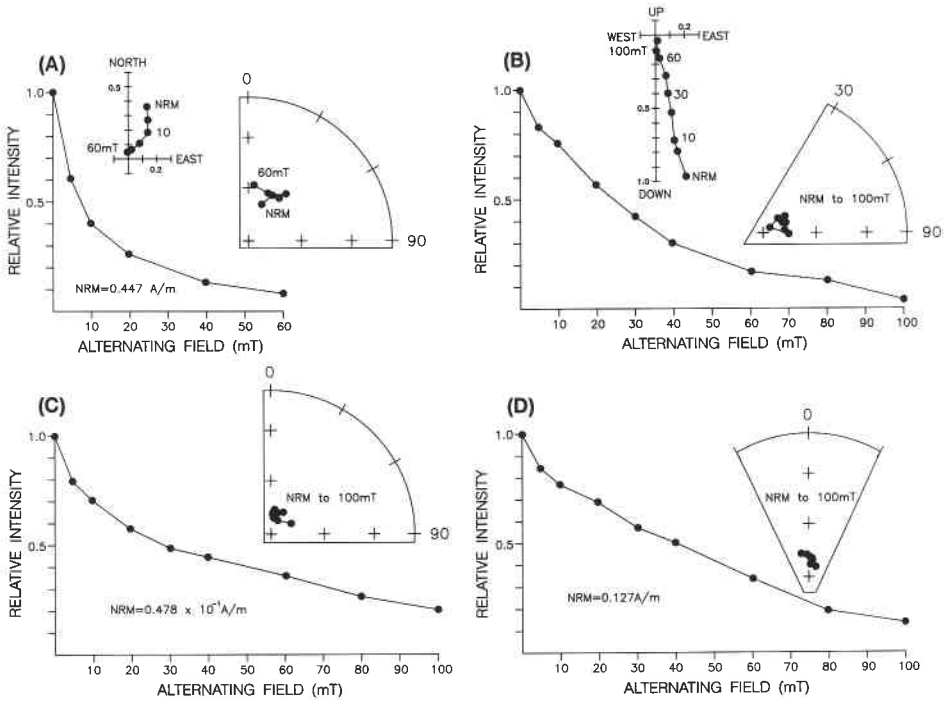


FIG. 4. Example behaviour of characteristic stable specimens upon AF step demagnetization: (A) Specimen is from site 5 with corresponding horizontal vector components also illustrated; (B) from site 14 with corresponding vertical vector components also illustrated; (C) from site 9; (D) from site 6. Diagrams illustrate decay of intensity with increasing AF's and the corresponding direction changes. All directions are normal polarity vectors plotted on the lower hemisphere of equal-area projections, which have been cropped.

reduced to <20% of their original NRM intensity, most often <10% (Fig. 4). This suggests that the principal magnetic carrier of these remanence components is probably magnetite, and observed T_{UB}'s support this conclusion. Examples of AF demagnetization in Fig. 4 illustrate the range of coercivity spectra observed as well as the stable, univectoral behaviour of specimens, from which stable end points and least squares lines could be readily obtained.

T_{UB} spectra are also variable but all show peak unblocking temperatures ≤570°C (Fig. 5). Above 570°C between 30–60% of the remanence components of all specimens was removed upon heating at 600°C, indicating that magnetite is the principal magnetic carrier of these remanence components. This result agrees with the observation of abundant accessory magnetite with virtually no hematite in the thin sections. Most specimens exhibited some distributed T_{UB} spectra, the range of which are shown by example specimens in Fig. 5; directions remained virtually unchanged. These demagnetization examples illustrate the characteristic, stable, univectoral behaviour of normal polarity specimens.

A summary of site mean data is given in Table 1. Core means assigned unit weight for the site mean calculations were usually obtained from two specimens. No data have been entered in the table for sites 2, 3, 19, and 20, as specimens from these sites failed to retain stable magnetic components. Sites 4 and 8 have been rejected owing to their large α_{95}'s >15°. Virtual geomagnetic poles (VGP's) from which the unit mean was obtained were calculated for each accepted site. Figure 6 illustrates the well-grouped core mean directions for the normal polarity directions isolated within the 14 accepted sites. Notable in Fig. 6 is the discordance between the observed unit mean direction (D = 026.2°, I = 73.5°, N = 14, k = 81, α_{95} = 4.4°) and the expected mid-Cretaceous direction (D = 332.5°, I = 75.1°) calculated from the mid-Cretaceous North American reference pole of Globerman and Irving (1988). The significant absence of reverse polarity directions, combined with the high T_{UB}'s and univectoral magnetic behaviour and the average of two biotite K–Ar ages of 88 Ma, indicates that this magnetization was acquired during the Cretaceous normal polarity superchron. The low dispersion of directions for this pluton (k = 81) can almost certainly be attributed to a time-integrated sampling of secular variation during cooling through its blocking temperature interval. The Spetch Creek pluton paleopole calculated from the site VGP's (Table 1) is listed in Table 2.

It is noteworthy that the observed Spetch Creek mean direction (D = 026.2°, I = 73.5°) is very close to the PEMF direction (D = 021°, I = 72°). It is important to point out that any potential PEMF overprints have likely been removed by the detailed AF and thermal step demagnetization procedures, combined with least squares principal component analysis of each specimen's demagnetization data. Vandall and Wheadon (1992) have shown these detailed procedures to be very effective methods of removing PEMF overprints. The example thermal demagnetization data in Fig. 5 illustrate the very stable high T_{UB} magnetizations that are observed for rocks of the Spetch Creek pluton, attesting to the geologic antiquity of it's magnetic remanence.

Geotectonics

Due to the virtual standstill in North American polar wander during the mid-Cretaceous, the reference pole is well constrained. Figure 2 illustrates the discordance between the Spetch Creek pluton pole (SC) and the mid-Cretaceous reference pole. This disagreement indicates that the Spetch Creek pluton

TABLE 2. Summary of Cretaceous Coast Belt data

Paleopole	Age (Ma)	B	λ (°N)	ϕ (°E)	A_{95} (°)	λ_p (°N)	RR (°)	RPD (°)	Reference
MS, Mount Stuart	<93	17	68	035	5.1*	27±4	−36±9	$32\pm6_a$	Beck et al. 1981
HS, Howe Sound	~95	17	83	129	11.1*	47±10	−17±16	$14\pm10_a$	Symons 1973*a*
CS, Eastern Panel	102−136	32	67	000	7.5*	37±6	−50±13	$30\pm7_a$	Symons 1977*b*
EB, Ecstall and Butedale	~70	15	74	323	10.0*	50±9	−43±18	$11\pm10_b$	Symons 1977*a*
SP, Spuzzum and Porteau	90−105	8	65	345	6.2	37±6	−57±11	$24\pm7_a$	Irving et al. 1985
SC, Spetch Creek	~88	14	73	282	7.0	59±7	−54±14	$3\pm7_a$	This study

NOTES: Age is from the original reference except for MS, where maximum age is from Walker and Brown (1991). B is the number of sampling sites; λ and ϕ are latitude and longitude of the paleopole; A_{95} is the confidence limit at $P = 0.05$ (* indicates an estimate of A_{95} from δp and δm semiaxes error ellipse); λ_p is the paleolatitude; RR (negative values indicate clockwise rotations) and RPD (positive values indicate northward displacement) are the relative rotations and relative paleolatitudinal displacements, respectively. ($_{a,b}$ correspond to the following reference poles used: 136−85 Ma, 71°N, 196°E, $A_{95} = 5°$, Globerman and Irving (1988), and 73−63 Ma, 78°N, 186°E, $A_{95} = 8°$, Marquis and Globerman (1988), respectively.

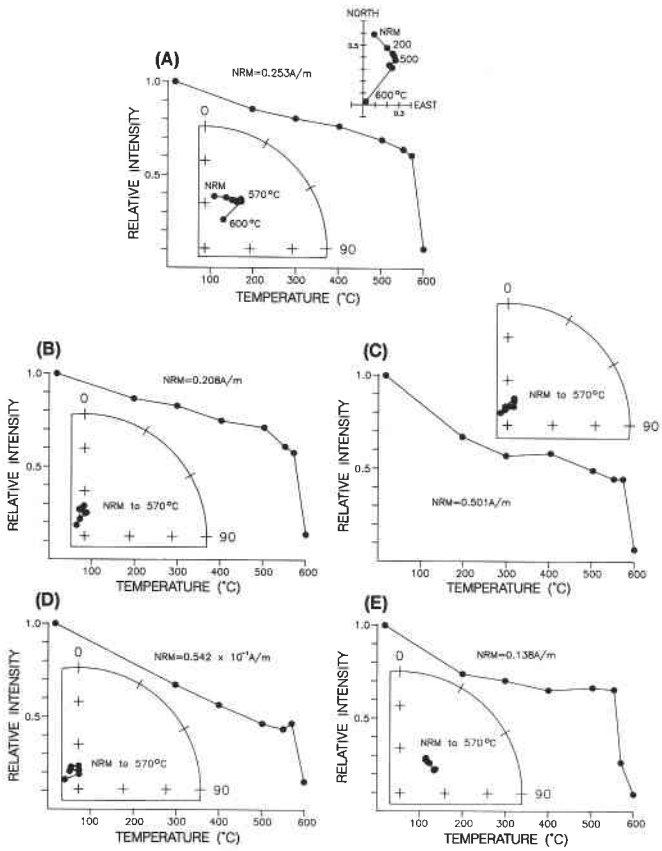


FIG. 5. Example behaviour of characteristic stable specimens upon thermal step demagnetization: (A) Specimen is from site 1 with corresponding horizontal vector components also illustrated; (B) from site 15; (C) from site 17; (D) from site 11; (E) from site 7. Diagrams illustrate decay of intensity with increasing temperature and the corresponding direction changes. Directions are plotted as in Fig. 4.

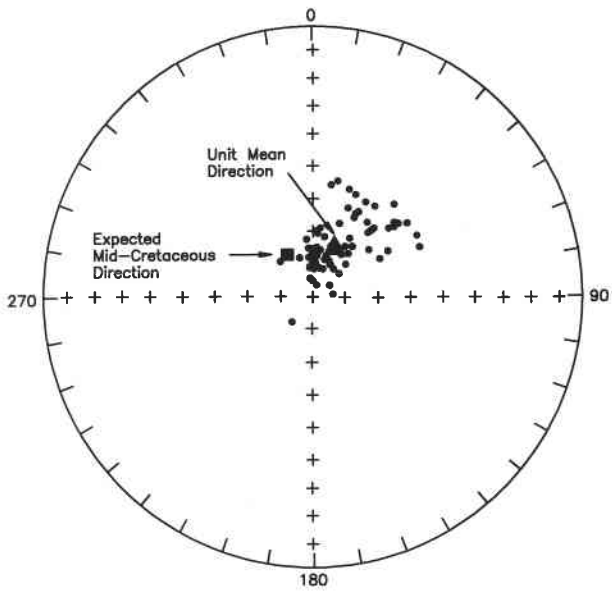


FIG. 6. Mean directions of 63 core magnetizations from 14 accepted collecting sites. Directions plotted as in Fig. 4.

has undoubtedly undergone tectonic tilt and (or) displacement and rotation since magnetization relative to stable North America.

If one accepts the tenet of Beck et al. (1981), Irving et al. (1985), and others that present horizontal for all the studied Cretaceous intrusives in the Coast Belt approximates paleohorizontal when they were magnetized, then the discordance of the Spetch Creek pole can be attributed to a northward RPD of 3 ± 7° and a clockwise RR of 54 ± 14°. This end member interpretation is inconsistent with the 2400 km of northward displacement invoked by the Baja British Columbia model given that the inferred northward displacement for the Spetch Creek pluton is only ~330 km. The maximum allowable northward displacement within error only approaches 1100 km with an attendant low probability (SC, Fig. 7). Adding the Spetch Creek pole (SC, Fig. 2) to the Cretaceous CB data base eliminates any perceived systematic discordance of the poles (e.g., Irving et al. 1985). The suggestion by Irving et al. (1985) that the Howe Sound data (Symons 1973*a*) represents a Tertiary overprint magnetization is highly speculative, as Vandall (1990) has pointed out. A recent paleomagnetic study of the Porteau pluton, published by Irving et al. (1992) in an abstract, suggests that the magnetizations observed by Symons (1973*a*) in Howe Sound were correctly interpreted. This follows given that Irving et al. (1992) outlined similar magnitudes of discordance (~1000 km) for the data from the Porteau pluton that are in agreement with the inferred displacement from the Howe Sound intrusives data (HS, Fig. 7). Thus the data from the CB including Spetch Creek and Howe

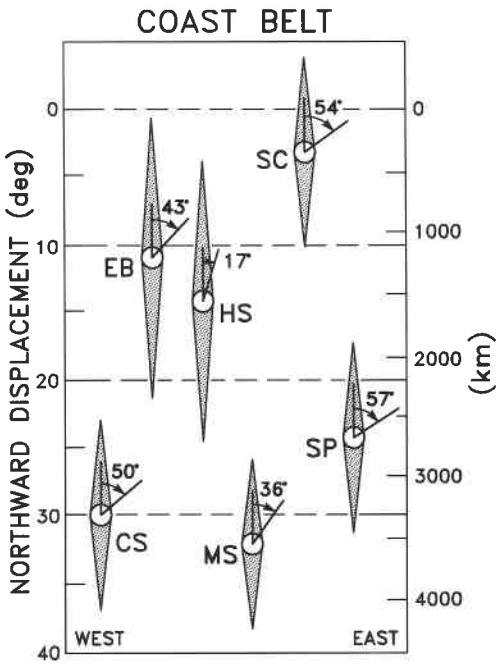


FIG. 7. Relative paleolatitudinal displacement and relative rotation diagram for Cretaceous Coast Belt data assuming present horizontal approximates paleohorizontal for all intrusives. Relative paleolatitudinal displacement error limits are shown with decreasing probability as stippled regions about the open circle mean displacement estimates. Relative rotation angles are outlined in degrees and symbols correspond to locations in Fig. 1 and poles in Fig. 2 and Table 2.

Sound preclude the discussion of a uniform discordance and Irving et al.'s (1992) discussion of tilt reaffirms Symons' (1973*a*) tilt hypothesis for Howe Sound.

Table 2 is a summary of the Cretaceous CB poles, paleolatitudes, and their corresponding relative displacement and rotation estimates assuming present horizontal approximates paleohorizontal. Figure 7 illustrates these data highlighting the variable magnitudes of inferred northward displacement ranging between 330 and 3500 km for the Spetch Creek pluton and Mount Stuart batholith, respectively. Clearly, the magnitudes of inferred displacement from these studies are not uniform, and therefore cannot be explained by the Baja British Columbia model, which invokes 2400 km of uniform displacement. Although the magnitudes of inferred displacements vary, they are consistently to the north (Fig. 7), resulting in the far-sidedness of the Cretaceous poles relative to the reference pole (Fig. 2). Inferred rotations also vary between 17 and 57° but they are consistently clockwise, resulting in the right-handedness of the Cretaceous poles relative to the reference pole (Fig. 2). In an east to west section across the CB the relative rotations decrease towards the centre and are largest nearest the easternmost and westernmost localities (Fig. 7). The significance of this observation is uncertain, as there are also large latitudinal differences between the locality positions and additional data would be required to confirm this trend. Is this related to boundary conditions on the margins of the CB? Although a uniform discordance of the poles is not observed, consistent northward relative displacements and clockwise relative rotations remain as striking features. It is very unlikely that present horizontal approximates paleohorizontal for all of the studied Cretaceous CB intrusives, as this assertion is inconsistent with the large variation observed in the discordance of their poles (Fig. 2). Thus, not only the geologic evidence as mentioned earlier (e.g., Symons 1973*a*, 1977*a*; Butler et al. 1989, 1990; Brown and Burmester 1991; Varsek et al. 1992) but also paleomagnetic evidence in the form of internal inconsistency support tilt as a significant cause of discordance for the Cretaceous CB data. Undoubtedly, the consistently northward relative displacements combined with the consistently clockwise rotations are the result of a common process.

Accepting that pluton tilting must be a significant source of the observed variation in the discordance of the Cretaceous CB data, a second end member interpretation of the Spetch Creek result is possible. By assuming present horizontal does not approximate paleohorizontal for the Spetch Creek pluton, an all-tilt model attributes the observed discordance to a simple tilt 15° down about an azimuth of 263°. The northeast-vergent thrusting between 91 and 86 Ma described previously provides a suitable syn- or postmagnetization mechanism that could have tilted the Spetch Creek pluton. An estimate of the magnitude and direction of tilt that is independent of the paleomagnetic data cannot yet be discerned from the available geologic information. However, the relatively small discordance of the Spetch Creek paleomagnetic direction is consistent with the interpreted low-angle east-northeast-vergent thrusts that are believed to underlie the Spetch Creek pluton. Recognizing that the geological constraints on paleohorizontal for the Spetch Creek pluton are very imprecise, a paleobarometric study in the area could provide some control. It is possible that the Spetch Creek pluton may have experienced a more complex history of concomitant tilt, displacement, and rotation. Certainly paleobarometric data from the Spuzzum pluton to the south suggests the possibility of a complex history, as does the poor agreement of CB displacements (Fig. 7).

Accepting the tenet of Symons (1973*a*, 1974, 1977*a*, 1977*b*) that his discordant data can be explained entirely by localized tilts, it can similarly be applied to the remaining Cretaceous CB data to arrive at an all-tilt model for the CB (Fig. 8; Table 3). Implicit in this hypothesis is that present horizontal is not the same as paleohorizontal. Tilting is assumed to be about a single horizontal axis. In this analysis observed directions are compared with expected directions calculated from the mid-Cretaceous reference pole and tilted about an appropriate horizontal axis to make the two directions coincide (Fig. 8). The downdip azimuth is at right angles to the horizontal tilt axis. The all-tilt model illustrates a variation in downdip azimuths from west to southwest and tilt angles between 12 and 33°. Unlike the displacement−rotation Baja British Columbia model, which requires uniform discordance due to en bloc translation and rotation of Baja British Columbia, tilt is accommodated in much smaller local-scale tectonic blocks with an implicit variation in direction and magnitude. Thus local block tilting is entirely consistent with the paleomagnetic observations, whereas uniform large-scale displacement (2400 km) and rotation are not. Consistent west to southwest down tilt suggests a common origin.

Not yet discussed is the extensive system of right-lateral strike-slip faults extending throughout the CC. Estimates of mid-Cretaceous to Eocene cumulative northward translation along these faults, located east of the CB, range between 500 and 900+ km (Gabrielse 1985; Price and Carmichael 1986). As this translation would have displaced terranes west of these structures, the CB likely moved 500−1000 km northward during this time, driven by Kula and (or) Farallon − North

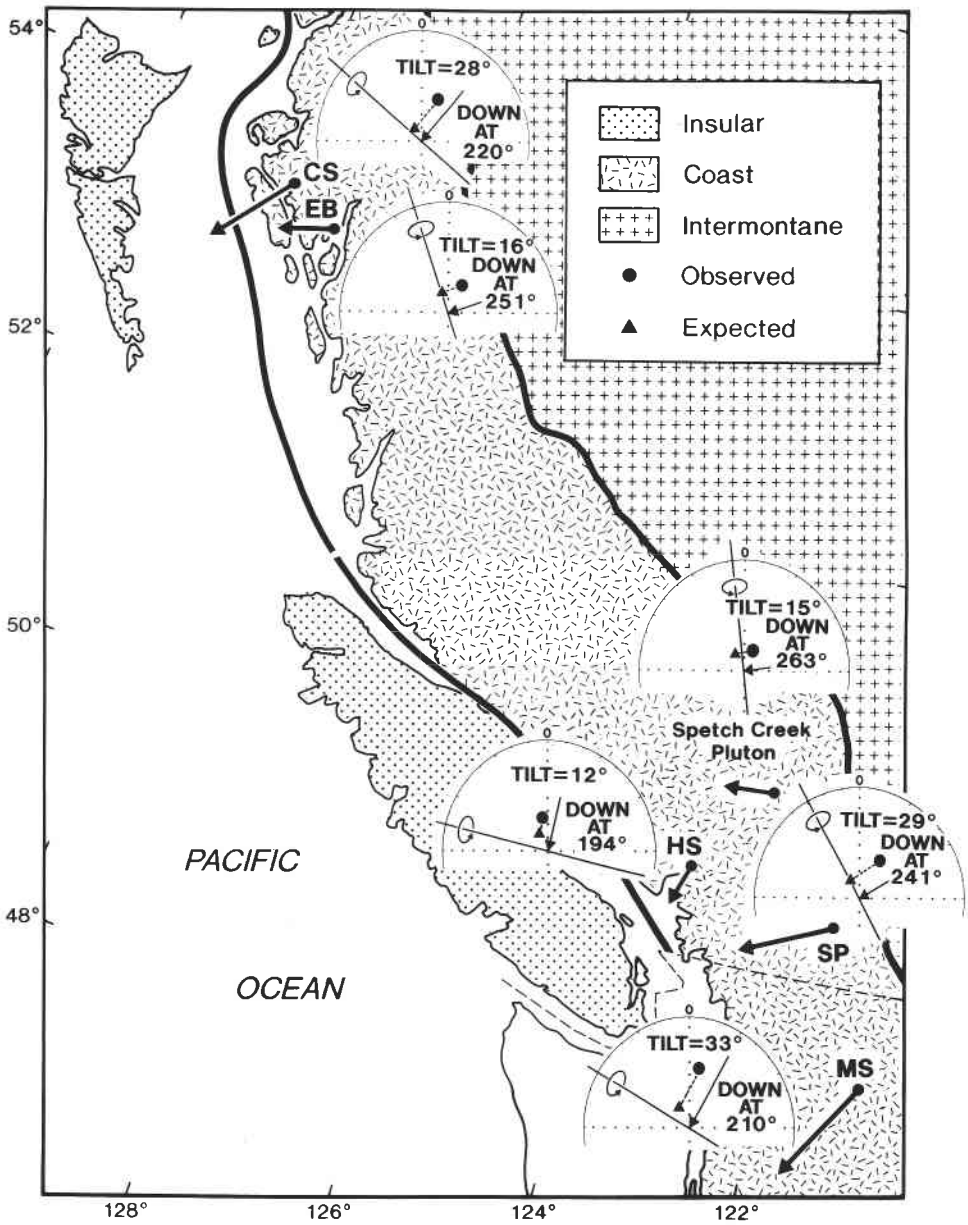


FIG. 8. All tilt scenarios for the Cretaceous Coast Belt data. Inset stereograms illustrate the downdip directions of tilt and the corresponding tilt magnitudes derived from comparison of the observed direction with the expected direction. Arrows at each locality illustrate downdip tilt direction in geographic map coordinates with their length proportional to the magnitude of tilt.

American plate interactions between 90 and 56 Ma (Engebretson et al. 1985). This motion and timing is consistent with orogen-parallel extension recognized in the northern Cascades and also believed to extend into the CB (Brown and Talbot 1989). It follows that the mid-Cretaceous intrusions of the CB underwent both northward displacement and tilting since the time of magnetization. Table 3 summarizes tectonic analyses for two combined tilt and displacement models invoking 500 and 1000 km of displacement. Observed CB directions are compared to expected cratonic reference directions that have been uniformly corrected for 500 km and then 1000 km of northward displacement. The subsequent tilt solution to reconcile the observed direction with a corrected expected direction yields the decoupled effects of combined displacement and tilt. A comparison of the data is highlighted in Table 3 where changes in the tilt angle and downdip azimuth are recorded as a result of either the combined "tilt + 500 km" or the combined "tilt + 1000 km" scenarios. Most notable is the minimal effect of either 500 or 1000 km of northward displacement on the tilt estimates for each intrusion. The effect of 500 km of northward displacement shallows the tilt angles by an average of only 1.3° and rotates the azimuths an average of 5.5° clockwise, overall a negligible effect. For 1000 km of northward displacement, the tilt angles shallow by an average of 2.5° and the azimuths rotate by an average of 13.6° clockwise, again a small effect. With increasing northward displacement the downdip azimuths consistently rotate clockwise towards the west and northwest. Thus, these displacements of geologically plausible magnitudes have a very small effect on the tilt scenarios, as Butler et al. (1990) suggested. This is strong evidence in support of tilt being responsible for much of the discordance of the Cretaceous CB poles, which substan-

TABLE 3. Tilt models for the Cretaceous Coast Belt data

Study	Model	T (°)	Az (°)	ΔT (°)	ΔAz (°)
MS, Mount Stuart	All Tilt	33	210		
	500 km + T	31	212	−2	2
	1000 km + T	28	216	−5	6
HS, Howe Sound	All Tilt	12	194		
	500 km + T	9	196	−3	2
	1000 km + T	7	209	−5	15
CS, Eastern Panel	All Tilt	28	220		
	500 km + T	27	228	−1	8
	1000 km + T	25	232	−3	12
EB, Ecstall and Butedale	All Tilt	16	251		
	500 km + T	15	259	−1	8
	1000 km + T	15	270	−1	19
SP, Spuzzum and Porteau	All Tilt	29	241		
	500 km + T	28	246	−1	5
	1000 km + T	27	249	−2	8
SC, Spetch Creek	All Tilt	15	263		
	500 km + T	15	271	0	8
	1000 km + T	16	285	+1	22
Average	500 km + T			−1.3	5.5
	1000 km + T			−2.5	13.6

NOTES: Model explanation in text. T is the tilt angle; Az is the downdip azimuth of tilt; ΔT is the change in the tilt angle due to the combined northward displacement (negative value corresponds to a decrease in the angle of dip from horizontal); ΔAz is the change in the downdip azimuth due to the combined northward displacement (positive value corresponds to a clockwise change in azimuth).

tiates the original tilt interpretations of Symons (e.g., 1973*a*, 1977*a*), with the exception of the inferred northward displacement. A number of local tilt mechanisms could be responsible for the tilt of any one of these Cretaceous CB intrusions, but given the consistent west-southwest down tilts, the most important are likely the recognized southwest–northeast CB compression, differential uplift, and extension (e.g., Atwater 1970; Parrish 1983; Gehrels and McClelland 1988; Rubin et al. 1990; Rusmore and Woodsworth 1991).

To avoid confusion the tilts discussed have been described as southwest-side-down consistent with the original authors' convention; however, Butler et al. (1989) have more appropriately described them as northeast-side-up. Although the two conventions are geometrically equivalent, the latter more accurately describes the actual tectonic process involving uplift. Thus, the northeast-side-up convention is used in the conclusions and abstract.

Conclusions

The mid-Cretaceous (~88 Ma) paleopole determined for the Spetch Creek pluton is discordant with the North American Cretaceous reference pole. This discordance can be explained by either displacement and rotation or tilt or some combination of these. If one invokes the displacement and rotation as in the Baja British Columbia model, which assumes present horizontal approximates paleohorizontal for all the Cretaceous CB intrusives, then the Spetch Creek pluton has been rotated 54° clockwise and displaced 330 ± 770 km northwards since the mid-Cretaceous. Alternatively, the Spetch Creek pluton may have been tilted 15° northeast-side-up. Regardless, the data from the Spetch Creek pluton provides a negative test of the Baja British Columbia model, which requires about 2400 km of northward displacement. Accepting the attendant uncertainties of paleohorizontal for the Spetch Creek pluton, this result reasonably limits maximum estimates of northward displacement to ~1000 km or somewhat less (SC, Fig. 7).

The review of the Cretaceous CB data including Spetch Creek demonstrates that the poles are not uniformly discordant. The wide variation in inferred relative displacements and rotations from the Cretaceous CB poles cannot be explained by the Baja British Columbia model. These discordant poles can be best explained by the tilt and moderate displacement (TMD) model, acknowledging (1) significant evidence of local tilting of plutons and (2) geologic evidence of moderate post-mid-Cretaceous to Eocene northward displacement (500–900+ km) of the amalgamated Insular, Coast, and Intermontane terranes. Paleomagnetic data from the CB is shown to be consistent with the model's moderate (500–1000 km) post-mid-Cretaceous northward displacement of amalgamated Insular, Coast, and Intermontane terranes west of the major dextral CC fault systems, combined with variable local block tilting up to the east and northeast. Tilt is shown to be the largest cause of discordance observed in Cretaceous CB data, broadly confirming Symons' (e.g., 1973*a*, 1977*a*) original conclusions. There is no evidence to support the previous assertion that present horizontal approximates paleohorizontal for all the studied Cretaceous CB plutons. Beyond the scope of this work are Cretaceous paleomagnetic data from the Intermontane Belt that also bear on the TMD model. A discussion of the agreement of these data with the TMD model is outlined by Vandall (1990).

This work emphasizes the need for much more CC paleomagnetic data in order to understand its complex tectonics. Paleomagnetic studies provide the best methods for detecting and quantifying tectonic displacements, rotations, and tilts in orogenic belts, yet independent geologic evidence is essential for the correct interpretation of the data. These tilt estimates

have assumed tilting about a single horizontal axis, although composite tilts are a distinct possibility. Studies of progressively older aged plutons, starting from a concordant Tertiary paleohorizontal reference datum within the same CB domain, may reveal a detailed tilt history for that region.

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